

Introduction to Linear Optimization and Extensions with MATLAB

by Roy H. Kwon

Chapter 1

Linear Programming

What is Linear Programming?

- Linear Programming is the field of optimization that concerns the
- (1) maximization or minimization of a linear function
- (2) subject to constraints that are also linear

Example: The Diet Problem

- Due to a limited budget you would like to find the most economical mix of food items subject to providing sufficient daily nutrition.
- The available food items are peanut butter, bananas, and chocolate.

Food Items and Nutritional Values

Food item	Fat (g/serving)	Carbohydrate (g/serving)	Protein (g/serving)
Peanut butter	130	51.6	64.7
Banana	1	51	2
Chocolate	12	22	2

Diet Problem: Cost of Food Items

Food Item	\$/serving
Peanut Butter	.20
Banana	.10
Chocolate	.15

Nutritional Requirements

- Suppose it is decided by a dietician that you need at least 35 grams of fat, at least 130 grams of carbohydrate, and at least 76 grams of protein daily.
- What is the least cost combination of the food items so that the nutritional requirements are satisfied?
- This problem can be formulated as a linear program.

Problem structure

- The problem has the following structure
- (1) there is an objective i.e. minimize cost of food items used
- (2) there are constraints i.e. food items must satisfy nutritional requirements
- Both of these constructs can be modelled mathematically through linear expressions in which case the problem is a linear program.

Problem Structure

Minimize cost of food items used

Subject to food items must provide enough fat

food items must provide enough carbohydrates

food items must provide enough protein

Decision Variables

Before the mathematical expressions can be created decision variables must be defined.

The decision variables for the diet problem represent the amount of each food item to be determined and are defined as

x_{pb} = servings of peanut butter

x_b = servings of bananas

x_c = servings of chocolate

Objective of the Diet Problem

The cost of a combination of the three food items is given by

$$.20x_{pb} + .10x_b + .15x_c$$

which is a linear expression since the variables are raised to the first power. This cost function is the objective function.

Then, the objective of the diet problem is to minimize this linear cost (objective function).

Constraints of the Diet Problem

The first constraint ensures that there is enough fat from the food items

$$130x_{pb} + 1x_b + 12x_c \geq 35$$

The second constraint ensures that there is enough carbohydrates

$$51.6x_{pb} + 51x_b + 22x_c \geq 130$$

The third constraint ensures that there is enough protein

$$64.7x_{pb} + 2x_b + 2x_c \geq 76$$

All constraints are linear expressions.

Non-negativity constraints

One last set of (linear) constraints involves imposing non-negativity constraints on the decision variables to ensure that the amounts determined make sense.

$$x_{pb}, x_b, x_c \geq 0$$

Diet Problem Linear Program

Putting together all of the parts the linear program can now be written as follows:

$$\text{minimize } .20x_{pb} + .10x_b + .15x_c$$

$$\text{subject to } 130x_{pb} + 1x_b + 12x_c \geq 35$$

$$51.6x_{pb} + 51x_b + 22x_c \geq 130$$

$$64.7x_{pb} + 2x_b + 2x_c \geq 76$$

$$x_{pb}, x_b, x_c \geq 0$$

General Diet Problem

- The diet problem can be generalized to consider n food items with m nutritional requirements

x_i = amount of food item i

c_i = cost of one serving of food item i

a_{ij} = amount of nutrient i in one serving of food item j

b_j = amount of nutrient j required

General Diet Problem

$$\text{minimize } c_1x_1 + c_2x_2 + \cdots + c_nx_n$$

$$\text{subject to } a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n \geq b_1$$

$$a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n \geq b_2$$

$$\vdots$$

$$a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n \geq b_m$$

$$x_1, x_2, \cdots, x_n \geq 0$$

Embedded Assumptions

- Proportionality
- Divisibility
- Additivity
- Certainty

Proportionality

- Proportionality means that the contribution towards the objective function and constraints of decisions are directly proportional to its values e.g. there are no economies of scale such as quantity based discounts.
- For example, in the diet problem every unit (serving) of peanut butter x_{pb} will contribute .20 towards the overall cost and 130 grams of fat towards the fat nutrient requirement.

Divisibility

- Divisibility means that the decision variables can take on any real number.
- For example, the amount of peanut butter in the diet problem can be less than one serving like 35% of a serving, i.e. $x_{pb} = .35$, or more than one serving.

Additivity

- Additivity means that the contribution of a decision variable towards the objective or constraints does not depend on other decision variables.
- In other words, the total contribution is the sum of individual contributions of each decision variable.
- For example, the contribution of every serving of peanut butter in the diet problem towards the overall fat requirement of 35 grams is 130 grams independent of the amount of servings of the other food items.

Certainty

- Certainty means that the data used as coefficients for a linear programming model such as the objective coefficients c and constraint coefficients A and b are known with certainty.

Steps for Linear Programming Model Formulation

- (1) Identify decision variables
- (2) Form objective function
- (3) Form constraints
- (4) Ensure all mathematical expressions are linear
- (5) Ensure all embedded assumptions are met

Production Planning

- A company produces n products.
- Each product uses m resources.
- Resources are limited i.e. there are only b_i units of resource i
- Each product j requires a_{ij} units of resource i
- Each unit of product j generates a revenue of p_j

Decision Variables

- Let X_j be the number (non-negative) of products of type j to be produced
- Then, the production plan is the vector

$$X = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_n \end{bmatrix}$$

Objective: Revenue Maximization

- The contribution towards revenue from the production of x_j units of product j is $p_j x_j$ and so the total revenue from a production plan x is then

$$p_1 x_1 + \cdots + p_n x_n = \sum_{j=1}^n p_j x_j$$

- which is a linear expression and this quantity should be maximized.

Constraints: Resource Limits

The production of x_j units of product j will consume $a_{ij}x_j$ units of resource i and so the total consumption of resource i by a production plan is

$$a_{i1}x_1 + a_{i2}x_2 + \cdots + a_{in}x_n = \sum_{j=1}^n a_{ij}x_j$$

Then, the total usage of each resource i must not exceed the amount available i.e.

$$\sum_{j=1}^n a_{ij}x_j \leq b_i, i = 1, \dots, m$$

Production Planning LP

$$\text{maximize } \sum_{j=1}^n p_j x_j$$

$$\text{subject to } \sum_{j=1}^n a_{ij} x_j \leq b_i \quad i = 1, \dots, m$$

$$x_j \geq 0 \quad j = 1, \dots, n$$

Financial Optimization: Bond Portfolio Construction

- Suppose that a bank has the following liability schedule

Year 1	Year 2	Year 3
\$12,000	\$18,000	\$20,000

- i.e. the bank needs to pay \$12,000 at the end of the first year, \$18,000 at the end of the second year, and \$20,000 at the end of the third year.

Bond Portfolio Construction

- Bonds are securities that are sold by agencies such as corporations or governments that entitle the buyer to periodic interest (coupon) payments and the payment of the principle (face value) at some time in the future (maturity).
- The bank wishes to use three bonds to form a portfolio (a collection of bonds) today to hold until all bonds have matured and that will generate the required cash to meet the liabilities.

Bond Portfolio Optimization

Bond	1	2	3
Price	102	99	98
Coupon	5	3.5	3.5
Maturity year	1	2	3

All bonds have face value of a \$100 and the coupons are annual (with one coupon per year).

For example, one unit of Bond 2 costs \$99 now and the holder will receive \$3.5 after 1 year and then \$3.5 plus the face value of \$100 at the end of the second year which is the maturity of Bond 2.

Decision Variables

Let the x_i = amount of bond i purchased.

Then, $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ is the bond portfolio.

Bond Portfolio Optimization

- (1) The objective is to minimize the cost of a portfolio
- (2) Each constraint should ensure that that cash flow generated by the bonds for a given time period is sufficient to match the liability for that period.

Bond Portfolio Linear Program

minimize $102x_1 + 99x_2 + 98x_3$

subject to $105x_1 + 3.5x_2 + 3.5x_3 \geq 12000$

$103.5x_2 + 3.5x_3 \geq 18000$

$103.5x_3 \geq 20000$

$x_1, x_2, x_3 \geq 0$

General Linear Programs

- The diet problem, production planning, and bond portfolio problems had constraints that were uniform in orientation e.g. The diet problem had all greater than or equal to constraints.
- Furthermore, the three examples also required all of the decision variables to be non-negative.
- In general, a linear program can be specified in many different ways the only major requirements being that the objective function and constraints have to be linear.

General Linear Programs

minimize or maximize $c^T x$

subject to

$$a_i^T x \leq b_i \quad i \in L$$

$$a_i^T x \geq b_i \quad i \in G$$

$$a_i^T x = b_i \quad i \in E$$

$$x_j \geq 0 \quad j \in NN$$

$$x_j \leq 0 \quad j \in NP$$

Example

maximize $5x_1 + x_2 - 3x_3$

subject to $x_1 + x_2 \leq 6$

$$x_2 + x_3 \geq 7$$

$$x_1 - x_3 = 2$$

$$x_1 \geq 0$$

$$x_2 \leq 0$$

x_3 unrestricted

Standard Form

minimize $c^T x$

subject to $Ax = b$

$x \geq 0$

Convert LP into Standard Form

1. Converting unrestricted variable

set $x = x^+ - x^-$, where $x^+ \geq 0$, $x^- \geq 0$

2. Converting inequality constraints

for a constraint i , $a_{i1}x_1 + a_{i2}x_2 + \cdots + a_{in}x_n \leq b_i$

add slack variable $s_i \geq 0$ so that

$$a_{i1}x_1 + a_{i2}x_2 + \cdots + a_{in}x_n + s_i = b_i$$

for a constraint i , $a_{i1}x_1 + a_{i2}x_2 + \cdots + a_{in}x_n \geq b_i$

add surplus variable $s_i \geq 0$ so that

$$a_{i1}x_1 + a_{i2}x_2 + \cdots + a_{in}x_n - s_i = b_i$$

3. Converting maximization to minimization

maximize $c^T x = -\text{minimize} -c^T x$

Example

maximize $5x_1 - 4x_2 + 6x_3$

subject to $-x_1 + x_2 \leq -7$

$$2x_2 - x_3 \geq 2$$

$$x_1 + 2x_3 = 7$$

$x_1 \geq 0, x_2 \geq 0, x_3$ unrestricted

$$\begin{aligned} x_3 &= x_3^+ - x_3^- \\ x_3^+ &\geq 0, x_3^- \geq 0 \end{aligned}$$

$$2x_2 - x_3 - s_2 = 2, s_2 \geq 0$$

$$\text{minimize } -(5x_1 - 4x_2 + 6x_3)$$

$$-x_1 + x_2 + s_1 = -7, s_1 \geq 0$$

$$\text{minimize } -5x_1 + 4x_2 - 6x_3^+ + 6x_3^-$$

$$\text{subject to } -x_1 + x_2 + s_1 = -7$$

$$2x_2 - x_3^+ + x_3^- - s_2 = 2$$

$$x_1 + 2x_3^+ - 2x_3^- = 7$$

$$x_1 \geq 0, x_2 \geq 0, x_3^+ \geq 0, x_3^- \geq 0,$$

$$s_1 \geq 0, s_2 \geq 0$$

Convert Minimization Problems with Absolute Value

$$\text{minimize } c_1 |x_1| + c_2 |x_2| + \cdots + c_n |x_n|$$

subject to

$$x_i \text{ unrestricted} \quad i = 1, \dots, n$$

- where $c_i > 0$ for all $i = 1, \dots, n$.
- Set $|x_i| = x_i^+ + x_i^-$, $x_i = x_i^+ - x_i^-$, $x_i^+, x_i^- \geq 0$, and submit into the problem, the above model then becomes

Covert Minimization Problems with Absolute Value

$$\text{minimize } c_1 (x_1^+ + x_1^-) + c_2 (x_2^+ + x_2^-) + \cdots + c_n (x_n^+ + x_n^-)$$

subject to

$$x_1 = x_1^+ - x_1^-$$

$$x_2 = x_2^+ - x_2^-$$

$$\vdots$$

$$x_n = x_n^+ - x_n^-$$

$$x_1^+ \geq 0, x_1^- \geq 0, x_2^+ \geq 0, x_2^- \geq 0, \dots, x_n^+ \geq 0, x_n^- \geq 0$$

Application: Portfolio Optimization

- Consider an investment in n stocks where each stock i has a random rate of return r_i with an expected return of u_i . Moreover, the covariance between the returns of stock i and stock j is σ_{ij} .
- Let x_i = the proportion of wealth invested in stock i , a portfolio is represented by the vector $x = (x_1, x_2, \dots, x_n)^T$.
- Markowitz (1952) construct a Mean-Variance Optimization (MVO) as follows:

Application: Portfolio Optimization

$$\text{minimize } \sum_{i=1}^n \sum_{j=1}^n \sigma_{ij} x_i x_j$$

$$\text{subject to } \sum_{i=1}^n u_i x_i = R$$

$$\sum_{i=1}^n x_i = 1$$

$$x_i \geq 0, \quad i = 1, \dots, n$$

- The objective is to minimize the risk of portfolio subject to meeting an expected goal R and ensuring the budget is exhausted.
- However the model is non-linear because of quadratic term in objective function and there are $n(n-1)/2$ covariance terms to estimate.

Application: Portfolio Optimization

- Konno and Yamazaki (1991) use the absolute deviation of the portfolio return instead of standard deviation, and construct Mean-Absolute Deviation (MAD) model.

$$\text{minimize } E \left| \sum_{i=1}^n (r_i - u_i) x_i \right| = \frac{1}{T} \sum_{t=1}^T \left| \sum_{i=1}^n (r_{it} - u_i) x_i \right|$$

$$\text{subject to } \sum_{i=1}^n u_i x_i = R$$

$$\sum_{i=1}^n x_i = 1$$

$$x_i \geq 0, \quad i = 1, \dots, n$$

- where r_{it} = the realized return for asset i at sub-period t .
- Set $\left| \sum_{i=1}^n (r_{it} - u_i) x_i \right| = s_t + y_t$, $\sum_{i=1}^n (r_{it} - u_i) x_i = s_t - y_t$, $s_t \geq 0$, $y_t \geq 0$, for all $t = 1, \dots, T$. The MAD becomes following LP:

Application: Portfolio Optimization

$$\text{minimize } \sum_{t=1}^T (s_t + y_t)$$

$$\text{subject to } \sum_{i=1}^n (r_{it} - u_i) x_i = s_t - y_t, \quad t = 1, \dots, T$$

$$\sum_{i=1}^n u_i x_i = R$$

$$\sum_{i=1}^n x_i = 1$$

$$x_i \geq 0, \quad i = 1, \dots, n$$

$$s_t \geq 0, y_t \geq 0, \quad t = 1, \dots, T$$

LP Terminology

- $A = m \times n$ matrix, the constraint coefficient.
- $c = n \times 1$ vector, the cost coefficient.
- $b = m \times 1$ vector, right hand side vector.
- $x = n \times 1$ vector, decision vector.
- $F = \{x \in R^n \mid Ax = b, x \geq 0\}$, feasible set.
- a vector $x^* \in F$ and $c^T x^* \leq c^T x$ for all $x \in F$, then x^* is the optimal solution of LP.
- LP is **bounded** if $L \leq c^T x$ for all $x \in F$ for some constant L , else the LP is said to be **unbounded**.

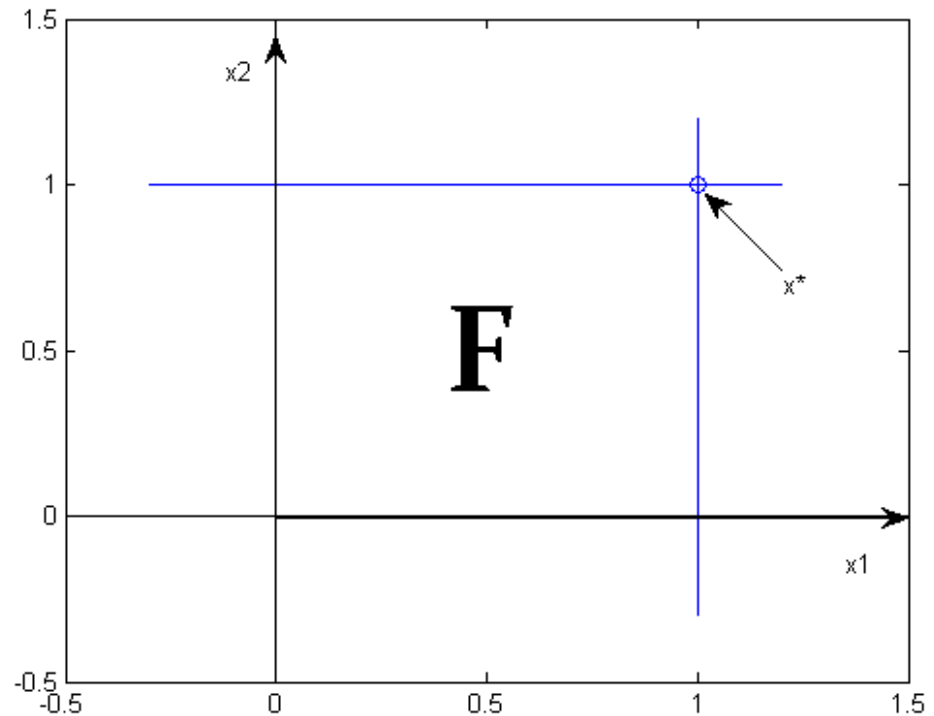
Example 1

minimize $-x_1 - x_2$

subject to $x_1 \leq 1$

$x_2 \leq 1$

$x_1 \geq 0, x_2 \geq 0$



$$F = \{x = (x_1, x_2)^T \in \mathbb{R}^2 \mid 0 \leq x_1 \leq 1, 0 \leq x_2 \leq 1\}$$

LP is bounded

$$x^* = (1, 1)^T$$

Example 1: Standard Form

minimize $-x_1 - x_2$

subject to $x_1 + x_3 = 1$

$$x_2 + x_4 = 1$$

$$x_1 \geq 0, x_2 \geq 0, x_3 \geq 0, x_4 \geq 0$$

$$\overline{F} = \left\{ x = (x_1, x_2, x_3, x_4)^T \in \mathbb{R}^4 \left| \begin{array}{l} x_1 + x_3 = 1, x_2 + x_4 = 1, \\ x_1 \geq 0, x_2 \geq 0, x_3 \geq 0, x_4 \geq 0 \end{array} \right. \right\}$$

LP is bounded since $x = (.5, .5, .5, .5)^T \in \overline{F}$

$$x^* = (1, 1, 0, 0)^T$$

Example 2: Unbounded

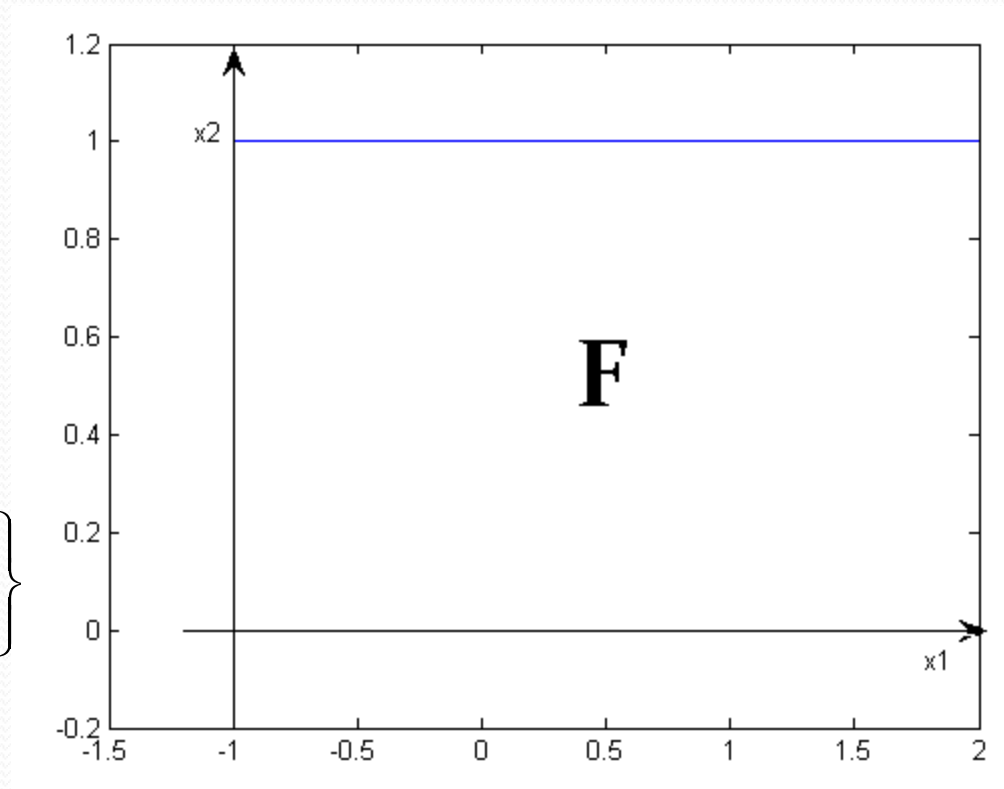
$$\begin{aligned} &\text{minimize } -x_1 - x_2 \\ &\text{subject to } -x_1 \leq 1 \\ &\quad \quad \quad x_2 \leq 1 \\ &\quad \quad \quad x_1 \geq 0, x_2 \geq 0 \end{aligned}$$

$$F = \left\{ x = (x_1, x_2)^T \in R^2 \left| \begin{array}{l} -x_1 \leq 1, x_2 \leq 1, \\ x_1 \geq 0, x_2 \geq 0 \end{array} \right. \right\}$$

LP is unbounded

$$x^{(k)} = \begin{bmatrix} k \\ 1 \end{bmatrix}, k = 1, 2, 3, \dots \text{ are all feasible solution,}$$

hence objective value $-k - 1 \rightarrow -\infty$ as $k \rightarrow \infty$



Network Optimization Models

- Many important applications can be stated as problems over networks. E.g. transportation problem.
- Networks can be used to represent many other problems where there is a notion of flow over an entity that can be represented as a network. E.g. the routing data over the Internet.

Network Optimization Models

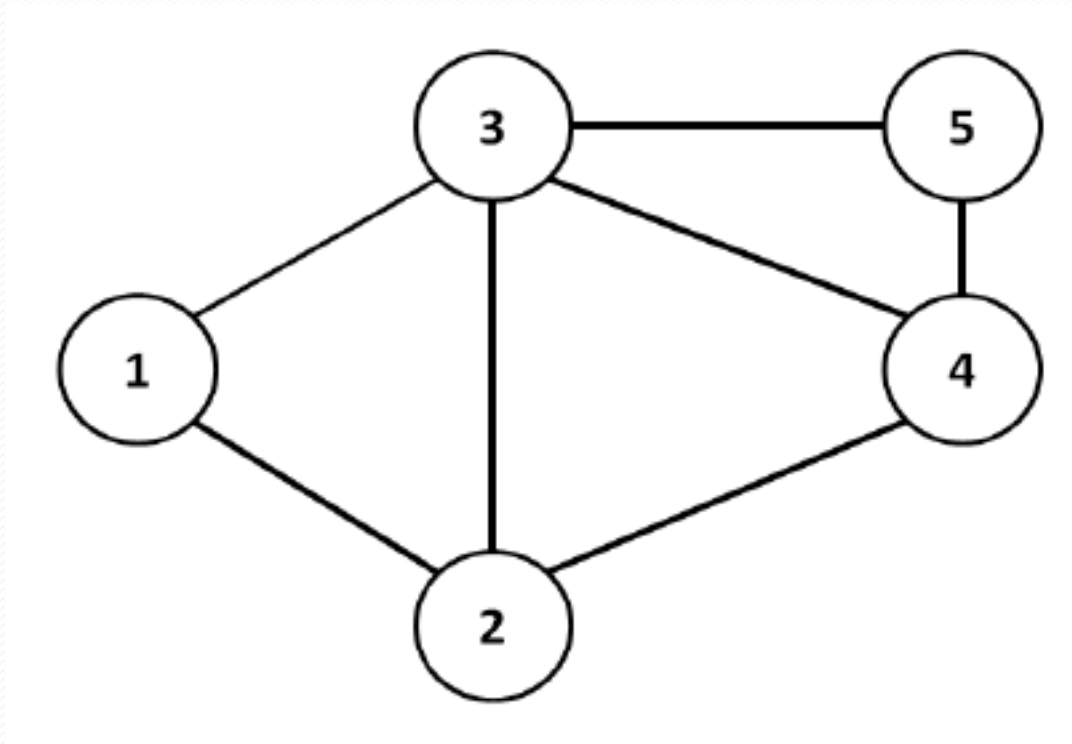
- Definition 1.12

A network (or graph) G consists of a set of nodes N and a set of edges E i.e. $G = (N, E)$.

G is an *undirected graph* when each edge $e \in E$ is an unordered pair of distinct nodes i and j in N . An edge between nodes i and j can be denoted in this case as $\{i, j\}$ or $\{j, i\}$.

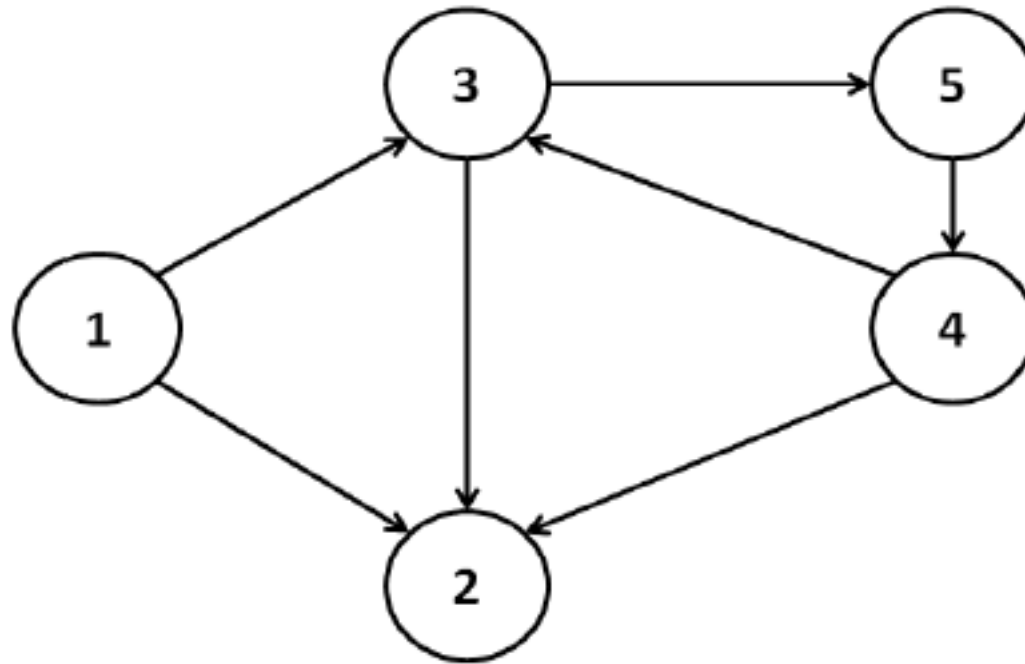
G is a *directed graph* when each edge $e \in E$ is an ordered pair of distinct nodes i and j in N . An edge will be denoted in this case as (i, j) and indicates that the direction of the edge is from node i to node j .

Example: Undirected Graph



$G = (N, E)$ where $N = \{1,2,3,4,5\}$ and
 $E = \{\{1,2\}, \{1,3\}, \{2,3\}, \{2,4\}, \{3,4\}, \{3,5\}, \{4,5\}\}$

Example: Directed Graph



$G = (N, E)$ where $N = \{1, 2, 3, 4, 5\}$ and
 $E = \{\{1, 2\}, \{1, 3\}, \{3, 2\}, \{4, 2\}, \{4, 3\}, \{3, 5\}, \{5, 4\}\}$

Network Optimization Models

- Definition 1.13

A path between nodes i and j of a graph G is a sequence of nodes and edges starting from i and ending with j such that no nodes are repeated. It is important to note that all edges specified in a path must be in E .

- Definition 1.14

We say that a graph $G = (N, A)$ is connected if there is a path between every pair of distinct nodes i and $j \in N$.

Minimum Cost Flow Problem

- Determine a least cost shipment (flow) of a product (item) through a connected directed network $G = (N, E)$ in order to satisfy demand at various nodes from various supply nodes.
- There are three types of nodes:
 - (1) supply node ($b_i > 0$)
 - (2) demand node ($b_i < 0$)
 - (3) transshipment nodes ($b_i = 0$)
- For each $(i, j) \in E$, c_{ij} is the cost of shipping one unit of product from node i to node j .

Minimum Cost Flow Problem

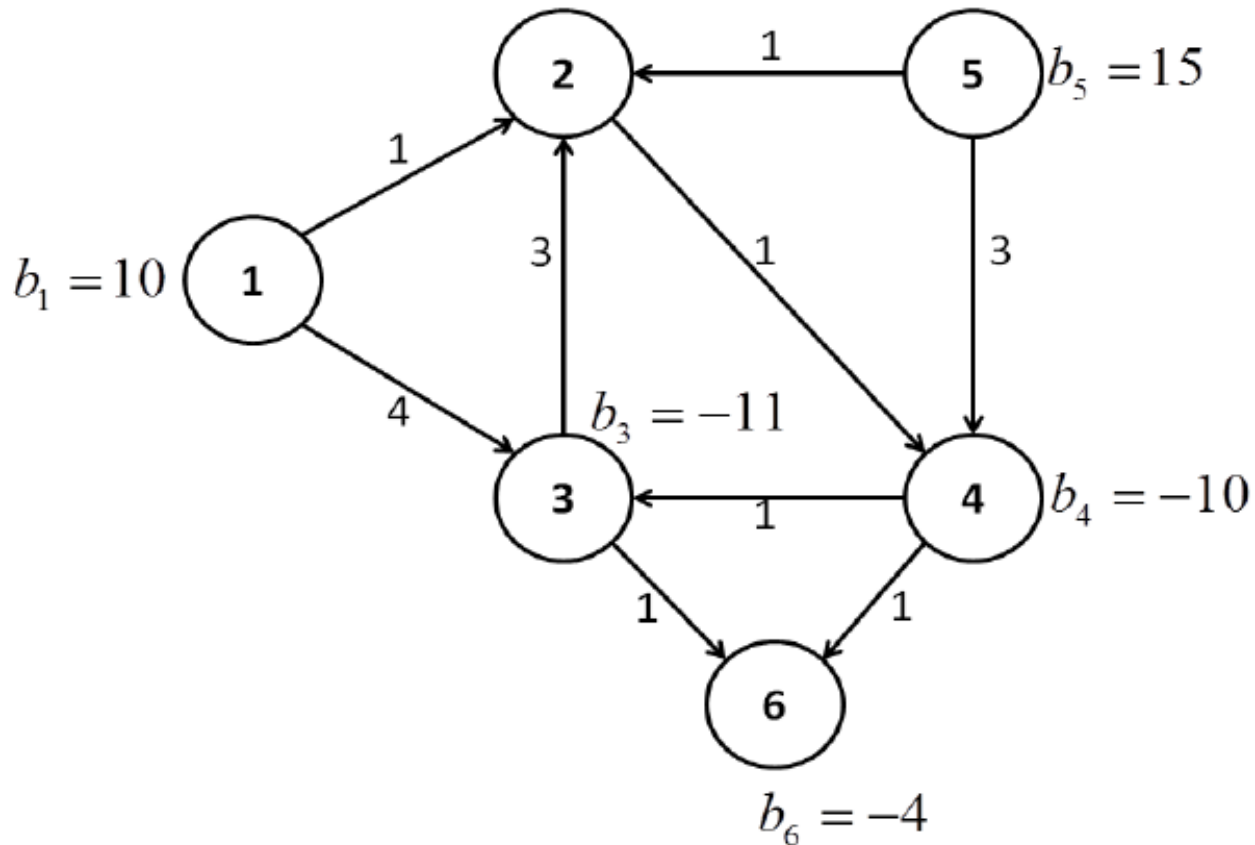
- Let x_{ij} = amount of product to be shipped from node i to node j .
- The minimum cost flow problem can be formulated as following LP:

$$\text{minimize } \sum_{(i,j) \in E} c_{ij} x_{ij}$$

$$\text{subject to } \sum_{\{j:(i,j) \in E\}} x_{ij} - \sum_{\{j:(j,i) \in E\}} x_{ji} = b_i \quad \text{for all } i \in N$$

$$l_{ij} \leq x_{ij} \leq u_{ij} \quad \text{for all } (i,j) \in E$$

Example: Minimum Cost Flow



Minimum cost flow network

Example: Minimum Cost Flow

- The problem can be formulated as LP:

$$\text{minimize } c^T x$$

$$\text{subject to } Ax = b$$

$$x \geq 0$$

$$x = \begin{bmatrix} x_{12} \\ x_{13} \\ x_{24} \\ x_{32} \\ x_{36} \\ x_{43} \\ x_{46} \\ x_{52} \\ x_{54} \end{bmatrix}, c = \begin{bmatrix} c_{12} \\ c_{13} \\ c_{24} \\ c_{32} \\ c_{36} \\ c_{43} \\ c_{46} \\ c_{52} \\ c_{54} \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \\ 1 \\ 3 \\ 1 \\ 1 \\ 1 \\ 1 \\ 3 \end{bmatrix}, b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \\ b_5 \\ b_6 \end{bmatrix} = \begin{bmatrix} 10 \\ 0 \\ -11 \\ -10 \\ 15 \\ -4 \end{bmatrix}, A = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & -1 & 0 & 0 & 0 & -1 & 0 \\ 0 & -1 & 0 & 1 & 1 & -1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 1 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & -1 & 0 & -1 & 0 & 0 \end{bmatrix}$$

Maximum Flow Problem

- Determine the maximum amount of flow that can be sent from a source node s to a sink node t over a directed network.
- There are three assumptions:
 - (1) The edge capacities are non-negative and integral.
 - (2) There is no directed path from s to t all of whose edges are uncapacitated (i.e. infinite capacity).
 - (3) There are no parallel edges.

Maximum Flow Problem

- Let v = the total flow out of the source

x_{ij} = amount of flow on edge $(i, j) \in E$.

- Then maximum flow problem can be formulated as following LP:

maximize v

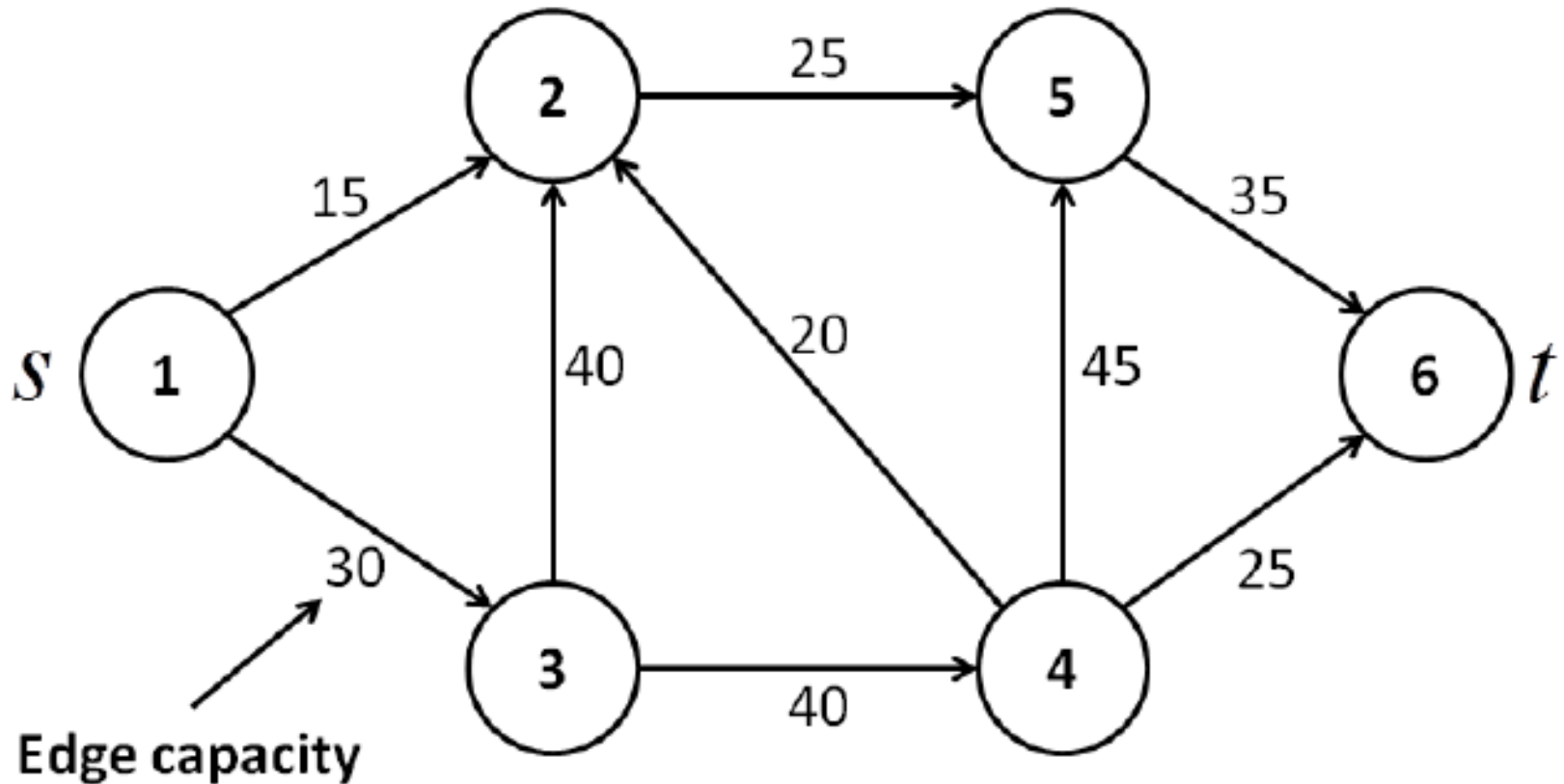
$$\text{subject to } \sum_{\{j:(i,j) \in E\}} x_{ij} - \sum_{\{j:(j,i) \in E\}} x_{ji} = v \quad \text{for } i = s$$

$$\sum_{\{j:(i,j) \in E\}} x_{ij} - \sum_{\{j:(j,i) \in E\}} x_{ji} = 0 \quad \text{for all } i \in N \text{ except } t \text{ and } s$$

$$\sum_{\{j:(i,j) \in E\}} x_{ij} - \sum_{\{j:(j,i) \in E\}} x_{ji} = -v \quad \text{for } i = t$$

$$0 \leq x_{ij} \leq u_{ij} \quad \text{for all } (i, j) \in E$$

Example: Maximum Flow Problem



Maximum flow network

Example: Maximum Flow Problem

- Maximize v

$$\text{Subject to } x_{12} + x_{13} = v$$

$$x_{25} - x_{12} - x_{32} - x_{42} = 0$$

$$x_{32} + x_{34} - x_{13} = 0$$

$$x_{42} + x_{45} + x_{46} - x_{34} = 0$$

$$x_{56} - x_{25} - x_{45} = 0$$

$$-x_{46} - x_{56} = -v$$

$$0 \leq x_{12} \leq 15, 0 \leq x_{13} \leq 30, 0 \leq x_{32} \leq 40,$$

$$0 \leq x_{25} \leq 25, 0 \leq x_{34} \leq 40, 0 \leq x_{42} \leq 20,$$

$$0 \leq x_{45} \leq 45, 0 \leq x_{56} \leq 35, 0 \leq x_{46} \leq 25$$

Solving LP by MATLAB

- The function *linprog* in MATLAB can be used to solve the following standard LP:

$$\text{minimize } f^T x$$

$$\text{subject to } Ax \leq b$$

$$Aeqx = beq$$

$$lb \leq x \leq ub$$



inequality constraint
equality constraint
lower and upper bound

- The optimal solution and objective value are obtained by the following statement:

$$[x, fval] = \text{linprog}(f, A, b, Aeq, beq, lb, ub)$$

Example 1

minimize $-3x_1 - 5x_2 - 3x_3$

subject to $x_1 - x_2 + x_3 \leq 15$

$2x_1 + 3x_2 + 6x_3 \leq 30$

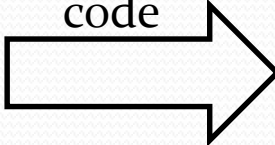
$2x_1 + x_2 = 30$

$0 \leq x_1 \leq 15$

$0 \leq x_2 \leq 10$

$0 \leq x_3 \leq 5$

Matlab
code



```
f = [-3, -5, -3]';  
A = [1, -1, 1;  
     2, 3, 6]; b = [15;  
                   30];  
Aeq = [2, 1, 0]; beq = [30];  
lb = [0, 0, 0]'; ub = [15, 10, 5]';  
[x, fval] = ...  
linprog(f, A, b, Aeq, beq, lb, ub);
```

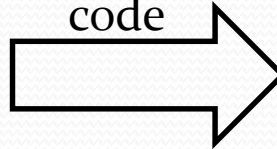


```
x =  
15.0000  
0.0000  
0.0000  
fval =  
-45.0000
```

Example 2

minimize $-3x_1 - 5x_2 - 3x_3$
subject to $x_1 - x_2 + x_3 \leq 15$
 $2x_1 + 3x_2 + 6x_3 \leq 30$
 $0 \leq x_1$
 $0 \leq x_2$
 $0 \leq x_3$

Matlab
code



```
f = [-3, -5, -3]';  
A = [1, -1, 1;  
     2, 3, 6]; b = [15;  
                   30];  
Aeq = [ ]; beq = [ ];  
lb = [0, 0, 0]';  
[x, fval] = ...  
    linprog(f, A, b, Aeq, beq, lb);
```



```
x =  
    0.0000  
   10.0000  
    0.0000  
fval =  
  -50.0000
```


Example 3

- Solve the minimum cost flow problem as follow:

```
>> f = [1;4;1;3;1;1;1;3];
```

```
>> Aeq = [1 1 0 0 0 0 0 0 0; -1 0 1 -1 0 0 0 -1 0;
```

```
0 -1 0 1 1 -1 0 0 0; 0 0 -1 0 0 1 1 0 -1;
```

```
0 0 0 0 0 0 0 1 1; 0 0 0 0 -1 0 -1 0 0];
```

```
>> beq = [10; 0; -11; -10; 15; -4];
```

```
>> A = []; b = []; lb = zeros(9,1); ub = [];
```

```
>> [x, fval]=linprog(f, A,b, Aeq,beq, lb,ub) ➔
```

x =
10.0000
0.0000
25.0000
0.0000
0.0000
11.0000
4.0000
15.0000
0.0000
fval =
65.0000